Thoughts And Rambles On Probability And Statistics

Problem Set 6: Convergence

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These are the end of chapter problems for Chapter 6 of the notes. Equation numbers refer to the full notes.

1. Let $X_1, X_2, \ldots$ be a sequence of random variables. Show that $X_n \xrightarrow{\text{qm}} b$ if and only if

$$\lim_{n\to\infty} \mathbb{E}(X_n) = b \quad \text{and} \quad \lim_{n\to\infty} \mathbb{V}(X_n) = 0$$

2. Let $X_1, X_2, \ldots$ be a sequence of random variables such that

$$\mathbb{P}\left(X_n = \frac{1}{n}\right) = 1 - \frac{1}{n^2} \quad \text{and} \quad \mathbb{P}(X_n = n) = \frac{1}{n^2}.$$

Does X_n converge in probability? Does X_n converge in quadratic mean?

3. Let $X_1, \ldots, X_n \sim \text{Bernoulli}(p)$. Prove that

$$\frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{\text{P}} p \quad \text{and} \quad \frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{\text{qm}} p.$$

4. Suppose that $X_n \sim N(0, 1/n)$ and let X be a random variable with distribution $F(x) = 0$ if $x < 0$ and $F(x) = 1$ if $x \geq 0$. Does X_n converge to X in probability? (Prove or disprove). Does X_n converge to X in distribution? (Prove or disprove).

5. Suppose that $X_n \in \text{Ge}(p_n), n \geq 1$. Show that

(a) if $p_n \to p > 0$, then $X_n \xrightarrow{d} \text{Ge}(p)$ as $n \to \infty$.

(b) if $p_n \to 0$ and $np_n \to \lambda > 0$, then X_n/n converges in distribution to an exponential with mean $1/\lambda$ as $n \to \infty$.

6 Let $X_1, X_2, \ldots$ be independent random variables with common characteristic function

$$\varphi(t) = \begin{cases} 1 - \sqrt{|t|(2-|t|)}, & \text{for } |t| \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

and let $\{S_n, n \geq 1\}$ denote the partial sums. Show that $\frac{S_n}{n^2}$ converges in distribution as

$n \to \infty$ and determine the limit.

7. Consider a sequence of random variables $\{X_n\}$ defined by:

$$X_n = \begin{cases} n & \text{with probability } \frac{1}{2n} \\ 0 & \text{with probability } 1 - \frac{1}{2n} \end{cases}$$

Is it uniformly integrable?